



US 20250282902A1

(19) **United States**

(12) **Patent Application Publication**  
**YAMAMOTO et al.**

(10) **Pub. No.: US 2025/0282902 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **FLUORINE-CONTAINING COPOLYMER**

Aug. 26, 2022 (JP) ..... 2022-135149

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Aug. 26, 2022 (JP) ..... 2022-135164

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**Publication Classification**

(51) **Int. Cl.**

**C08F 214/26** (2006.01)

**C09D 5/03** (2006.01)

**C09D 127/18** (2006.01)

**H01B 3/44** (2006.01)

(52) **U.S. Cl.**

CPC ..... **C08F 214/262** (2013.01); **C08F 214/26**  
(2013.01); **C09D 5/03** (2013.01); **C09D**  
**127/18** (2013.01); **H01B 3/445** (2013.01);  
**C08F 2800/20** (2013.01)

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(21) Appl. No.: **18/895,509**

(22) Filed: **Sep. 25, 2024**

(57)

**ABSTRACT**

**Related U.S. Application Data**

(63) Continuation of application No. PCT/JP2023/  
013252, filed on Mar. 30, 2023.

There is provided a fluorine-containing copolymer comprising tetrafluoroethylene unit, hexafluoropropylene unit, and a fluoro(alkyl vinyl ether) unit, wherein the copolymer has a content of hexafluoropropylene unit of 9.7 to 10.3% by mass with respect to the whole of the monomer units, a content of the fluoro(alkyl vinyl ether) unit of 0 to 0.3% by mass with respect to the whole of the monomer units, a melt flow rate at 372° C. of 0.80 to 1.30 g/10 min, and a total number of a carbonyl group-containing terminal group, —CF=CF<sub>2</sub> and —CH<sub>2</sub>OH per 10<sup>6</sup> main-chain carbon atoms of 50 or less.

**Foreign Application Priority Data**

(30)

Mar. 30, 2022 (JP) ..... 2022-055227

Mar. 30, 2022 (JP) ..... 2022-055233

Jul. 15, 2022 (JP) ..... 2022-114217

Aug. 26, 2022 (JP) ..... 2022-135142

## FLUORINE-CONTAINING COPOLYMER

## CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a Rule 53(b) Continuation of International Application No. PCT/JP2023/013252 filed Mar. 30, 2023, which claims priorities based on Japanese Patent Application No. 2022-055227 filed Mar. 30, 2022, Japanese Patent Application No. 2022-055233 filed Mar. 30, 2022, Japanese Patent Application No. 2022-114217 filed Jul. 15, 2022, Japanese Patent Application No. 2022-135164 filed Aug. 26, 2022, Japanese Patent Application No. 2022-135142 filed Aug. 26, 2022, and Japanese Patent Application No. 2022-135149 filed Aug. 26, 2022, the respective disclosures of which are incorporated herein by reference in their entirety.

## TECHNICAL FIELD

[0002] The present disclosure relates to a fluorine-containing copolymer.

## BACKGROUND ART

[0003] Patent Literature 1 describes a method for producing a tetrafluoroethylene-hexafluoropropylene copolymer characterized in that tetrafluoroethylene and hexafluoropropylene are copolymerized in an aqueous medium in the presence of a di(fluoroacyl) peroxide as a polymerization initiator and methanol or ethanol as a molecular weight regulator.

## RELATED ART

## Patent Literature

[0004] Patent Literature 1: Japanese Patent Publication No. 59-02821

## SUMMARY

[0005] According to the present disclosure, there is provided a fluorine-containing copolymer comprising tetrafluoroethylene unit, hexafluoropropylene unit and a fluoro(alkyl vinyl ether) unit, wherein the copolymer has a content of hexafluoropropylene unit of 9.7 to 10.3% by mass with respect to the whole of the monomer units, a content of the fluoro(alkyl vinyl ether) unit of 0 to 0.3% by mass with respect to the whole of the monomer units, a melt flow rate at 372° C. of 0.80 to 1.30 g/10 min, and the total number of a carbonyl group-containing terminal group,  $\text{—CF=CF}_2$  and  $\text{—CH}_2\text{OH}$  per 10<sup>6</sup> main-chain carbon atoms of 50 or less.

## Effects

[0006] According to the present disclosure, there can be provided a fluorine-containing copolymer being hardly deformed even in a melt state which can give a formed article being excellent in the 45° C. abrasion resistance, the 82.5° C. high-temperature rigidity, the 115° C. tensile creep resistance, the 50° C. creep resistance, the durability to repeated loads, the chemical solution low permeation, and the ductility to tensile force applied at 100° C.

## DESCRIPTION OF EMBODIMENTS

[0007] Hereinafter, specific embodiments of the present disclosure will be described in detail, but the present disclosure is not limited to the following embodiments.

[0008] The fluorine-containing copolymer of the present disclosure comprises tetrafluoroethylene (TFE) unit, hexafluoropropylene (HFP) unit, and optionally, a fluoro(alkyl vinyl ether) (FAVE) unit.

[0009] Fluororesins, which are excellent in the chemical solution resistance and the heat resistance, are used as a material for forming chemical solution piping. Although as such fluororesins, TFE/HFP copolymers (FEP), TFE/perfluoro(propyl vinyl ether) copolymers (PFA), and the like are known, for example, in the case of using FEP described in Patent Literature 1, in the case where a relatively high-temperature chemical solution is made to flow in piping, these copolymers have a problem of abrasion of the piping. Further, there are cases where pipes and tubes formed from FEP are stretched into desired shapes, but in the case of using FEP described in Patent Literature 1, it cannot be said that the ductility is excellent, and an improvement is demanded.

[0010] It has been found that by regulating the contents of HFP unit and the FAVE unit, the melt flow rate and the number of functional groups of the fluorine-containing copolymer containing TFE unit, HFP unit and the FAVE unit, in very limited ranges, the 82.5° C. high-temperature rigidity, the 115° C. tensile creep resistance, the 50° C. creep resistance, the durability to repeated loads, and the chemical solution low permeation (low permeation to chemical solutions such as acetic acid or ammonia water) of formed articles obtained from the fluorine-containing copolymer are improved, and simultaneously, the 45° C. abrasion resistance, and the ductility to tensile force applied at 100° C. are also remarkably improved.

[0011] Further, since the fluorine-containing copolymer of the present disclosure is small in the own weight deformation even in the melt state, even in the case of being formed into thick pipes, the obtained pipes have clean cross-sections and also have uniform thicknesses.

[0012] The fluorine-containing copolymer of the present disclosure is a melt-fabricable fluororesin. Being melt-fabricable means that a polymer can be melted and processed by using a conventional processing device such as an extruder or an injection molding machine.

[0013] An FAVE constituting the FAVE unit includes at least one selected from the group consisting of monomers represented by the general formula (1):



wherein Y<sup>1</sup> denotes F or CF<sub>3</sub>; Rf denotes a perfluoroalkyl group having 1 to 5 carbon atoms; and p denotes an integer of 0 to 5, and q denotes an integer of 0 to 5, and monomers represented by the general formula (2):



wherein X are identical or different and each denote H, F or CF<sub>3</sub>; and R<sup>1</sup> includes a linear or branched fluoroalkyl group having 1 to 6 carbon atoms and optionally containing one or two of at least one atom selected from the group consisting of H, Cl, Br and I, or a cyclic fluoroalkyl group having 5 or 6 carbon atoms and optionally containing one or two of at least one atom selected from the group consisting of H, Cl, Br and I.

**[0014]** Among these, the FAVE is preferably a monomer represented by the general formula (1), more preferably at least one selected from the group consisting of perfluoro (methyl vinyl ether), perfluoro(ethyl vinyl ether) (PEVE) and perfluoro(propyl vinyl ether) (PPVE), still more preferably at least one selected from the group consisting of PEVE and PPVE and especially preferably PPVE.

**[0015]** The content of HFP unit of the fluorine-containing copolymer is, with respect to the whole of the monomer units, 9.7 to 10.3% by mass, and preferably 9.8% by mass or higher and preferably 10.2% by mass or lower. When the content of HFP unit is too low, formed articles excellent in the 45° C. abrasion resistance and the ductility to tensile force applied at 100° C. cannot be obtained. When HFP unit is too high, formed articles excellent in the 82.5° C. high-temperature rigidity, the 115° C. tensile creep resistance, the 50° C. creep resistance, and the durability to repeated loads cannot be obtained.

**[0016]** The content of the FAVE unit of the fluorine-containing copolymer is, with respect to the whole of the monomer units, 0 to 0.3% by mass, and preferably 0.2% by mass or lower and more preferably 0.1% by mass or lower. Due to that the content of the FAVE unit is in the above range, formed articles excellent in the 45° C. abrasion resistance, the 82.5° C. high-temperature rigidity, the 115° C. tensile creep resistance, the 50° C. creep resistance, the durability to repeated loads, the chemical solution low permeation, and the ductility to tensile force applied at 100° C. can be obtained. When the content of the FAVE unit is too high, formed articles excellent in the 82.5° C. high-temperature rigidity, the 115° C. tensile creep resistance, and the 50° C. creep resistance cannot be obtained.

**[0017]** The content of TFE unit of the fluorine-containing copolymer is, with respect to the whole of the monomer units, preferably 89.4% by mass or higher, more preferably 89.5% by mass or higher, still more preferably 89.6% by mass or higher and further still more preferably 89.7% by mass or higher, and preferably 90.3% by mass or lower. Then, the content of TFE unit may be selected so that the total of contents of HFP unit, the FAVE unit, TFE unit and other monomer units becomes 100% by mass.

**[0018]** The fluorine-containing copolymer of the present disclosure is not limited as long as the copolymer contains the above three monomer units, and may be a copolymer containing only the above three monomer units, or may be a copolymer containing the above three monomer units and other monomer units.

**[0019]** The other monomers are not limited as long as being copolymerizable with TFE, HFP and FAVE, and may be fluoromonomers or fluorine-non-containing monomers.

**[0020]** It is preferable that the fluoromonomer is at least one selected from the group consisting of chlorotrifluoroethylene, vinyl fluoride, vinylidene fluoride, trifluoroethylene, hexafluoroisobutylene, monomers represented by  $\text{CH}_2=\text{CZ}^1(\text{CF}_2)_n\text{Z}^2$  (wherein  $\text{Z}^1$  is H or F,  $\text{Z}^2$  is H, F or Cl, and  $n$  is an integer of 1 to 10), perfluoro(alkyl vinyl ether)s [PAVE] represented by  $\text{CF}_2=\text{CFO}-\text{CH}_2-\text{Rf}^2$  (wherein  $\text{Rf}^2$  is a perfluoroalkyl group having 1 to 5 carbon atoms), perfluoro-2,2-dimethyl-1,3-dioxol [PDD], and perfluoro-2-methylene-4-methyl-1,3-dioxolane [PMD].

**[0021]** The monomers represented by  $\text{CH}_2=\text{CZ}^1(\text{CF}_2)_n\text{Z}^2$  include  $\text{CH}_2=\text{CFCF}_3$ ,  $\text{CH}_2=\text{CH}-\text{C}_4\text{F}_9$ ,  $\text{CH}_2=\text{CH}-\text{C}_6\text{F}_{13}$ , and  $\text{CH}_2=\text{CF}-\text{C}_3\text{F}_6\text{H}$ .

**[0022]** The fluorine-non-containing monomers include hydrocarbon-based monomers copolymerizable with TFE, HFP and FAVE. Examples of the hydrocarbon-based monomers include alkenes such as ethylene, propylene, butylene, and isobutylene; alkyl vinyl ethers such as ethyl vinyl ether, propyl vinyl ether, butyl vinyl ether, isobutyl vinyl ether, and cyclohexyl vinyl ether; vinyl esters such as vinyl acetate, vinyl propionate, n-vinyl butyrate, vinyl isobutyrate, vinyl valerate, vinyl pivalate, vinyl caproate, vinyl caprylate, vinyl caprate, vinyl versate, vinyl laurate, vinyl myristate, vinyl palmitate, vinyl stearate, vinyl benzoate, vinyl para-t-butylbenzoate, vinyl cyclohexanecarboxylate, vinyl monochloroacetate, vinyl adipate, vinyl acrylate, vinyl methacrylate, vinyl crotonate, vinyl sorbate, vinyl cinnamate, vinyl undecylenate, vinyl hydroxyacetate, vinyl hydroxypropionate, vinyl hydroxybutyrate, vinyl hydroxyvalerate, vinyl hydroxyisobutyrate, and vinyl hydroxycyclohexanecarboxylate; alkyl allyl ethers such as ethyl allyl ether, propyl allyl ether, butyl allyl ether, isobutyl allyl ether, and cyclohexyl allyl ether; and alkyl allyl esters such as ethyl allyl ester, propyl allyl ester, butyl allyl ester, isobutyl allyl ester, and cyclohexyl allyl ester.

**[0023]** The fluorine-non-containing monomers may also be functional group-containing hydrocarbon-based monomers copolymerizable with TFE, HFP and FAVE. Examples of the functional group-containing hydrocarbon-based monomers include hydroxyalkyl vinyl ethers such as hydroxyethyl vinyl ether, hydroxypropyl vinyl ether, hydroxybutyl vinyl ether, hydroxyisobutyl vinyl ether, and hydroxycyclohexyl vinyl ether; fluorine-non-containing monomers having a glycidyl group, such as glycidyl vinyl ether and glycidyl allyl ether; fluorine-non-containing monomers having an amino group, such as aminoalkyl vinyl ethers and aminoalkyl allyl ethers; fluorine-non-containing monomers having an amido group, such as (meth)acrylamide and methylolacrylamide; bromine-containing olefins, iodine-containing olefins, bromine-containing vinyl ethers, and iodine-containing vinyl ethers; and fluorine-non-containing monomers having a nitrile group.

**[0024]** The content of the other monomer units in the fluorine-containing copolymer of the present disclosure is, with respect to the whole of the monomer units, preferably 0 to 0.9% by mass, and more preferably 0.7% by mass or lower, still more preferably 0.5% by mass or lower and especially preferably 0.1% by mass or lower.

**[0025]** The melt flow rate (MFR) of the fluorine-containing copolymer is 0.80 to 1.30 g/10 min. The MFR of the copolymer is preferably 0.85 g/10 min or higher, more preferably 0.90 g/10 min or higher, still more preferably 1.00 g/10 min or higher, especially preferably 1.06 g/10 min or higher and most preferably 1.10 g/10 min or higher, and preferably 1.29 g/10 min or lower, more preferably 1.28 g/10 min or lower, still more preferably 1.19 g/10 min or lower, especially preferably 1.15 g/10 min or lower and most preferably 1.10 g/10 min or lower. When the MFR is too low, formed articles excellent in the 82.5° C. high-temperature rigidity cannot be obtained. Further, when the MFR is too low, formed articles excellent in the ammonia low permeation cannot be obtained; and by regulating the MFR in the above range, the chemical solution low permeation including the ammonia low permeation can be improved. When the MFR is too high, formed articles excellent in the 45° C. abrasion resistance, and the ductility to tensile force applied at 100° C. cannot be obtained. Further, when the

MFR is too high, the fluorine-containing copolymer becomes one which is easily deformed in the melt state, and it becomes difficult to obtain uniform-thickness thick pipes and the like.

[0026] In the present disclosure, the MFR is a value obtained as a mass (g/10 min) of a polymer flowing out from a die of 2 mm in inner diameter and 8 mm in length per 10 min at 372° C. under a load of 5 kg using a melt indexer G-01 (manufactured by Toyo Seiki Seisaku-sho Ltd.), according to ASTM D1238.

[0027] The MFR can be regulated by regulating the kind and amount of a polymerization initiator to be used in polymerization of monomers, the kind and amount of a chain transfer agent, and the like.

[0028] The fluorine-containing copolymer of the present disclosure may or may not have a carbonyl group-containing terminal group,  $-\text{CF}=\text{CF}_2$  or  $-\text{CH}_2\text{OH}$ . In the fluorine-containing copolymer of the present disclosure, the total number of the carbonyl group-containing terminal group,  $-\text{CF}=\text{CF}_2$  and  $-\text{CH}_2\text{OH}$  is, per  $10^6$  main-chain carbon atoms, 50 or less, preferably 40 or less, more preferably 30 or less, still more preferably 20 or less and especially preferably 15 or less. By making the total number of the carbonyl group-containing terminal group,  $-\text{CF}=\text{CF}_2$  and  $-\text{CH}_2\text{OH}$  in the above range, formed articles excellent in the chemical solution low permeation (low permeation to chemical solutions such as acetic acid, particularly acetic acid low permeation) can be obtained. The total number of the carbonyl group-containing terminal group,  $-\text{CF}=\text{CF}_2$  and  $-\text{CH}_2\text{OH}$  can be regulated, for example, by suitable selection of the kind of a polymerization initiator or a chain transfer agent, or by a wet heat treatment or fluorination treatment of the fluorine-containing copolymer described later.

[0029] The carbonyl group-containing terminal groups are for example,  $-\text{COF}$ ,  $-\text{COOH}$ ,  $-\text{COOR}$  (R is an alkyl group),  $-\text{CONH}_2$ , and  $-\text{O}(\text{C}=\text{O})\text{O}-\text{R}$  (R is an alkyl group). The kinds of the alkyl groups (R)  $-\text{COOR}$  and  $-\text{O}(\text{C}=\text{O})\text{O}-\text{R}$  have are determined depending on a polymerization initiator or a chain transfer agent used in

resistance of the fluorine-containing copolymer becomes excellent. The number of  $-\text{CF}_2\text{H}$  of the fluorine-containing copolymer, per  $10^6$  main-chain carbon atoms, may be 50 or more, and is preferably 60 or more, more preferably more than 70, still more preferably more than 80, especially preferably 90 or more and most preferably 100 or more. The upper limit of the number of  $-\text{CF}_2\text{H}$  is not limited, and may be, for example, 800. The number of  $-\text{CF}_2\text{H}$  can be regulated, for example, by suitable selection of the kind of a polymerization initiator or a chain transfer agent, or by a wet heat treatment or fluorination treatment of the fluorine-containing copolymer described later.

[0031] For identification of the kind of the functional groups and measurement of the number of the functional groups, infrared spectroscopy can be used.

[0032] The number of the functional groups is measured, specifically, by the following method. First, the fluorine-containing copolymer is molded by cold press to prepare a film of 0.25 to 0.30 mm in thickness. The film is analyzed by Fourier transform infrared spectroscopy to obtain an infrared absorption spectrum, and a difference spectrum against a base spectrum that is completely fluorinated and has no functional groups is obtained. From an absorption peak of a specific functional group observed on this difference spectrum, the number N of the functional group per  $1 \times 10^6$  carbon atoms in the fluorine-containing copolymer is calculated according to the following formula (A).

$$N = I \times K / t \quad (\text{A})$$

[0033] I: absorbance

[0034] K: correction factor

[0035] t: thickness of film (mm)

[0036] For reference, for some functional groups, the absorption frequency, the molar absorption coefficient and the correction factor are shown in Table 1. Then, the molar absorption coefficients are those determined from FT-IR measurement data of low molecular model compounds.

TABLE 1

| Functional Group                         | Absorption Frequency (cm <sup>-1</sup> ) | Molar Extinction Coefficient (l/cm/mol) | Correction Factor | Model Compound                                           |
|------------------------------------------|------------------------------------------|-----------------------------------------|-------------------|----------------------------------------------------------|
| $-\text{COF}$                            | 1883                                     | 600                                     | 388               | $\text{C}_7\text{F}_{15}\text{COF}$                      |
| $-\text{COOH}$ free                      | 1815                                     | 530                                     | 439               | $\text{H}(\text{CF}_2)_6\text{COOH}$                     |
| $-\text{COOH}$ bonded                    | 1779                                     | 530                                     | 439               | $\text{H}(\text{CF}_2)_6\text{COOH}$                     |
| $-\text{COOCH}_3$                        | 1795                                     | 680                                     | 342               | $\text{C}_7\text{F}_{15}\text{COOCH}_3$                  |
| $-\text{CONH}_2$                         | 3436                                     | 506                                     | 460               | $\text{C}_7\text{H}_{15}\text{CONH}_2$                   |
| $-\text{CH}_2\text{OH}_2$ , $-\text{OH}$ | 3648                                     | 104                                     | 2236              | $\text{C}_7\text{H}_{15}\text{CH}_2\text{OH}$            |
| $-\text{CF}_2\text{H}$                   | 3020                                     | 8.8                                     | 26485             | $\text{H}(\text{CF}_2\text{CF}_2)_3\text{CH}_2\text{OH}$ |
| $-\text{CF}=\text{CF}_2$                 | 1795                                     | 635                                     | 366               | $\text{CF}_2=\text{CF}_2$                                |

production of the fluorine-containing copolymer, and are, for example, alkyl groups having 1 to 6 carbon atoms, such as  $-\text{CH}_3$ .

[0030] The fluorine-containing copolymer of the present disclosure may or may not have  $-\text{CF}_2\text{H}$ . It is preferable that the fluorine-containing copolymer of the present disclosure has  $-\text{CF}_2\text{H}$ , because in the case of melt forming the fluorine-containing copolymer, it becomes difficult for forming defects such as foaming to be caused, and the heat

[0037] Absorption frequencies of  $-\text{CH}_2\text{CF}_2\text{H}$ ,  $-\text{CH}_2\text{COF}$ ,  $-\text{CH}_2\text{COOH}$ ,  $-\text{CH}_2\text{COOCH}_3$  and  $-\text{CH}_2\text{CONH}_2$  are lower by a few tens of kayzers (cm<sup>-1</sup>) than those of  $-\text{CF}_2\text{H}$ ,  $-\text{COF}$ ,  $-\text{COOH}$  free and  $-\text{COOH}$  bonded,  $-\text{COOCH}_3$  and  $-\text{CONH}_2$  shown in the Table, respectively.

[0038] For example, the number of the functional group  $-\text{COF}$  is the total of the number of a functional group determined from an absorption peak having an absorption

frequency of  $1,883\text{ cm}^{-1}$  derived from  $\text{—CF}_2\text{COF}$  and the number of a functional group determined from an absorption peak having an absorption frequency of  $1,840\text{ cm}^{-1}$  derived from  $\text{—CH}_2\text{COF}$ .

**[0039]** The number of  $\text{—CF}_2\text{H}$  groups can also be determined from a peak integrated value of the  $\text{—CF}_2\text{H}$  group acquired in a  $^{19}\text{F}$ -NMR measurement using a nuclear magnetic resonance spectrometer and set at a measurement temperature of (the melting point of a polymer+20) $^\circ\text{C}$ .

**[0040]** Functional groups such as a  $\text{—CF}_2\text{H}$  group are functional groups present on the main chain terminals or side chain terminals of the fluorine-containing copolymer, and functional groups present on the main chain or the side chains thereof. These functional groups are introduced to the fluorine-containing copolymer, for example, by a chain transfer agent or a polymerization initiator used in production of the fluorine-containing copolymer. For example, in the case of using an alcohol as the chain transfer agent, or a peroxide having a structure of  $\text{—CH}_2\text{OH}$  as the polymerization initiator,  $\text{—CH}_2\text{OH}$  is introduced on the main chain terminals of the fluorine-containing copolymer. Alternatively, the functional group is introduced on the side chain terminal of the fluorine-containing copolymer by polymerizing a monomer having the functional group.

**[0041]** By carrying out a treatment such as a wet heat treatment or a fluorination treatment on the fluorine-containing copolymer having such functional groups, there can be obtained the fluorine-containing copolymer having the number of functional groups in the above range. The fluorine-containing copolymer is more preferably one having been subjected to a wet heat treatment.

**[0042]** The melting point of the fluorine-containing copolymer is preferably  $220$  to  $290^\circ\text{C}$ . and more preferably  $260$  to  $280^\circ\text{C}$ . Due to that the melting point is in the above range, the copolymer which is hardly deformed even in a melt state and the formed article which is excellent in the  $45^\circ\text{C}$ . abrasion resistance, the  $82.5^\circ\text{C}$ . high-temperature rigidity, the  $115^\circ\text{C}$ . tensile creep resistance, the  $50^\circ\text{C}$ . creep resistance, the durability to repeated loads, the chemical solution low permeation, and the ductility to tensile force applied at  $100^\circ\text{C}$ . can be obtained.

**[0043]** In the present disclosure, the melting point can be measured by using a differential scanning calorimeter [DSC].

**[0044]** The acetic acid permeability of the copolymer is preferably  $3.0\text{ (g}\cdot\text{cm/m}^2\text{)}$  or lower and more preferably  $2.7\text{ (g}\cdot\text{cm/m}^2\text{)}$  or lower. The copolymer of the present disclosure has excellent acetic acid low permeation due to suitably regulated contents of HFP unit and the FAVE unit, melt flow rate (MFR) and number of functional groups. That is, by using the copolymer of the present disclosure, formed articles which chemical solutions such as acetic acid hardly permeates can be obtained.

**[0045]** In the present disclosure, the acetic acid permeability can be measured under the condition of a temperature of  $60^\circ\text{C}$ . and for 55 days. Specific measurement of the acetic acid permeability can be carried out by a method described in Examples.

**[0046]** The ammonia water permeability of the copolymer is preferably  $800\text{ mg}\cdot\text{cm/m}^2$  or lower. The copolymer of the present disclosure has excellent ammonia low permeation due to suitably regulated contents of HFP unit and the FAVE unit, melt flow rate (MFR) and number of functional groups. That is, by using the copolymer of the present disclosure,

formed articles which chemical solutions such as ammonia water hardly permeates can be obtained.

**[0047]** In the present disclosure, the ammonia water permeability can be measured under the condition of a temperature of  $37^\circ\text{C}$ . and for 28 days. Specific measurement of the ammonia water permeability can be carried out by a method described in Examples.

**[0048]** The fluorine-containing copolymer of the present disclosure can be produced by any polymerization method of bulk polymerization, suspension polymerization, solution polymerization, emulsion polymerization and the like. In these polymerization methods, conditions such as temperature and pressure, a polymerization initiator, a chain transfer agent, a solvent and other additives can suitably be set depending on the composition and the amount of a desired fluorine-containing copolymer.

**[0049]** The polymerization initiator may be an oil-soluble radical polymerization initiator or a water-soluble radical initiator.

**[0050]** An oil-soluble radical polymerization initiator may be a known oil-soluble peroxide, and examples thereof typically include:

**[0051]** dialkyl peroxydicarbonates such as di-n-propyl peroxydicarbonate, diisopropyl peroxydicarbonate, di-sec-butyl peroxydicarbonate;

**[0052]** peroxyesters such as t-butyl peroxyisobutyrate and t-butyl peroxyvalerate;

**[0053]** dialkyl peroxides such as di-t-butyl peroxide; and

**[0054]** di[fluoro(or fluorochloro)acyl]peroxides.

**[0055]** The di[fluoro(or fluorochloro)acyl]peroxides include diacyl peroxides represented by  $[(\text{RfCOO})\text{—}]_2$  wherein Rf is a perfluoroalkyl group, an  $\omega$ -hydroperfluoroalkyl group or a fluorochloroalkyl group.

**[0056]** Examples of the di[fluoro(or fluorochloro)acyl]peroxides include di( $\omega$ -hydroperfluorohexanoyl) peroxide, di( $\omega$ -hydro-dodecafluoroheptanoyl) peroxide, di( $\omega$ -hydro-tetradecafluorooctanoyl) peroxide, di( $\omega$ -hydrohexadecafluorononanoyl) peroxide, di(perfluorobutyryl) peroxide, di(perfluorovaleryl) peroxide, di(perfluorohexanoyl) peroxide, di(perfluoroheptanoyl) peroxide, di(perfluorooctanoyl) peroxide, di(perfluorononanoyl) peroxide, di( $\omega$ -chlorohexafluorobutyryl) peroxide, di( $\omega$ -chlorodecafluorohexanoyl) peroxide, di( $\omega$ -chlorotetradecafluorooctanoyl) peroxide,  $\omega$ -hydrododecafluoroheptanoyl- $\omega$ -hydrohexadecafluorononanoyl peroxide,  $\omega$ -chloro-hexafluorobutyryl- $\omega$ -chloro-decafluorohexanoyl peroxide,  $\omega$ -hydrododecafluoroheptanoyl-perfluorobutyryl peroxide, di(dichloropentafluorobutanoyl) peroxide, di(trichlorooctafluorohexanoyl) peroxide, di(tetrachloroundecafluorooctanoyl) peroxide, di(pentachlorotetradecafluorodecanoyl) peroxide and di(undecachlorotriacontafluorodocosanoyl) peroxide.

**[0057]** The water-soluble radical polymerization initiator may be a well known water-soluble peroxide, and examples thereof include ammonium salts, potassium salts and sodium salts of persulfuric acid, perboric acid, perchloric acid, perphosphoric acid, percarbonic acid and the like, and t-butyl permaleate and t-butyl hydroperoxide. A reductant such as a sulfite salt may be combined with a peroxide and used, and the amount thereof to be used may be 0.1 to 20 times with respect to the peroxide.

**[0058]** Use of an oil-soluble radical polymerization initiator as the polymerization initiator is preferable, because of

enabling avoidance of the formation of —COF and —COOH, and enabling easy regulation of the total number of —COF and —COOH of the fluorine-containing copolymer in the above-mentioned range. Further, use of the oil-soluble radical polymerization initiator is likely to make it easy also for the carbonyl group-containing terminal group and —CH<sub>2</sub>OH to be regulated in the above-mentioned range. It is especially suitable that the fluorine-containing copolymer is produced by suspension polymerization using the oil-soluble radical polymerization initiator. The oil-soluble radical polymerization initiator is preferably at least one selected from the group consisting of dialkyl peroxy-carbonates and di[fluoro(or fluorochloro)acyl]peroxides, and more preferably at least one selected from the group consisting of di-n-propyl peroxydicarbonate, diisopropyl peroxydicarbonate, and di( $\omega$ -hydro-dodecafluoroheptanoyl) peroxide.

**[0059]** Examples of the chain transfer agent include hydrocarbons such as ethane, isopentane, n-hexane and cyclohexane; aromatics such as toluene and xylene; ketones such as acetone; acetates such as ethyl acetate and butyl acetate; alcohols such as methanol, ethanol and 2,2,2-trifluoroethanol; mercaptans such as methyl mercaptan; halogenated hydrocarbons such as carbon tetrachloride, chloroform, methylene chloride and methyl chloride; and 3-fluorobenzotrifluoride. The amount thereof to be added can vary depending on the magnitude of the chain transfer constant of a compound to be used, but the chain transfer agent is used usually in the range of 0.01 to 20 parts by mass with respect to 100 parts by mass of a solvent.

**[0060]** For example, in the cases of using a dialkyl peroxy-carbonate, a di[fluoro(or fluorochloro)acyl]peroxide or the like as a polymerization initiator, although there are some cases where the molecular weight of an obtained fluorine-containing copolymer becomes too high and the regulation of the melt flow rate to a desired one is not easy, the molecular weight can be regulated by using the chain transfer agent. It is especially suitable that the fluorine-containing copolymer is produced by suspension polymerization using the chain transfer agent such as an alcohol and the oil-soluble radical polymerization initiator.

**[0061]** The solvent includes water and mixed solvents of water and an alcohol. A monomer to be used for the polymerization of the fluorine-containing copolymer of the present disclosure can also be used as the solvent.

**[0062]** In the suspension polymerization, in addition to water, a fluorosolvent may be used. The fluorosolvent may include hydrochlorofluoroalkanes such as CH<sub>3</sub>CClF<sub>2</sub>, CH<sub>3</sub>CCl<sub>2</sub>F, CF<sub>3</sub>CF<sub>2</sub>CCl<sub>2</sub>H and CF<sub>2</sub>ClCF<sub>2</sub>CFHCl; chlorofluoroalkanes such as CF<sub>2</sub>ClCF<sub>2</sub>CF<sub>2</sub>CF<sub>3</sub> and CF<sub>3</sub>CF<sub>2</sub>ClCF<sub>2</sub>CF<sub>3</sub>; and perfluoroalkanes such as perfluorocyclobutane, CF<sub>3</sub>CF<sub>2</sub>CF<sub>2</sub>CF<sub>3</sub>, CF<sub>3</sub>CF<sub>2</sub>CF<sub>2</sub>CF<sub>2</sub>CF<sub>3</sub> and CF<sub>3</sub>CF<sub>2</sub>CF<sub>2</sub>CF<sub>2</sub>CF<sub>2</sub>CF<sub>3</sub>, and among these, perfluoroalkanes are preferred. The amount of the fluorosolvent to be used is, from the viewpoint of the suspensibility and the economic efficiency, preferably 10 to 100 parts by mass with respect to 100 parts by mass of the solvent.

**[0063]** The polymerization temperature is not limited, and may be 0 to 100° C. In the case where the decomposition rate of the polymerization initiator is too high, including cases of using a dialkyl peroxy-carbonate, a di[fluoro(or fluorochloro)acyl]peroxide or the like as the polymerization initiator, it is preferable to adopt a relatively low polymerization temperature such as in the temperature range of 0 to 35° C.

**[0064]** The polymerization pressure can suitably be determined according to other polymerization conditions such as the kind of the solvent to be used, the amount of the solvent, the vapor pressure and the polymerization temperature, but usually may be 0 to 9.8 MPaG. The polymerization pressure is preferably 0.1 to 5 MPaG, more preferably 0.5 to 2 MPaG and still more preferably 0.5 to 1.5 MPaG. When the polymerization pressure is 1.5 MPaG or higher, the production efficiency can be improved.

**[0065]** Examples of the additives in the polymerization include suspension stabilizers. The suspension stabilizers are not limited as long as being conventionally well-known ones, and methylcellulose, polyvinyl alcohols and the like can be used. With the use of a suspension stabilizer, suspended particles produced by the polymerization reaction are dispersed stably in an aqueous medium, and therefore the suspended particles hardly adhere on the reaction vessel even when a SUS-made reaction vessel not having been subjected to adhesion preventing treatment such as glass lining is used. Accordingly, a reaction vessel withstanding a high pressure can be used, and therefore the polymerization under a high pressure becomes possible and the production efficiency can be improved. By contrast, in the case of carrying out the polymerization without using the suspension stabilizer, the suspended particles may adhere and the production efficiency may be lowered with the use of a SUS-made reaction vessel not having been subjected to adhesion preventing treatment is used. The concentration of the suspension stabilizer in the aqueous medium can suitably be regulated depending on conditions.

**[0066]** In the case of obtaining an aqueous dispersion containing a fluoropolymer by a polymerization reaction, a dried fluoropolymer may be recovered by coagulating, cleaning and drying the fluorine-containing copolymer contained in the aqueous dispersion. Alternatively, in the case of obtaining the fluorine-containing copolymer as a slurry by a polymerization reaction, a dried fluoropolymer may be recovered by taking out the slurry from a reaction vessel, and cleaning and drying the slurry. The fluorine-containing copolymer can be recovered in a powder form by the drying.

**[0067]** The fluorine-containing copolymer obtained by the polymerization may be formed into pellets. A method of forming into pellets is not limited, and a conventionally known method can be used. Examples thereof include methods of melt extruding the fluorine-containing copolymer by using a single-screw extruder, a twin-screw extruder or a tandem extruder and cutting the resultant into a predetermined length to form the fluorine-containing copolymer into pellets. The extrusion temperature in the melt extrusion needs to be varied depending on the melt viscosity and the production method of the fluorine-containing copolymer, and is preferably the melting point of the fluorine-containing copolymer+20° C. to the melting point of the fluorine-containing copolymer+140° C. A method of cutting the fluorine-containing copolymer is not limited, and there can be adopted a conventionally known method such as a strand cut method, a hot cut method, an underwater cut method, or a sheet cut method. Volatile components in the obtained pellets may be removed by heating the pellets (degassing treatment). Alternatively, the obtained pellets may be treated by bringing the pellets into contact with hot water of 30 to 200° C., steam of 100 to 200° C. or hot air of 40 to 200° C.

**[0068]** The fluorine-containing copolymer obtained by the polymerization may be heated in the presence of air and

water at a temperature of 100° C. or higher (wet heat treatment). Examples of the wet heat treatment include a method in which by using an extruder, the fluorine-containing copolymer obtained by the polymerization is melted and extruded while air and water are fed. The wet heat treatment can convert thermally unstable functional groups of the fluorine-containing copolymer, such as —COF and —COOH, to thermally relatively stable —CF<sub>2</sub>H, whereby the total number of —COF and —COOH and the total number of carbonyl group-containing terminal groups and —CH<sub>2</sub>OH of the fluorine-containing copolymer can easily be regulated in the above-mentioned ranges. By heating the fluorine-containing copolymer, in addition to air and water, in the presence of an alkali metal salt, the conversion reaction to —CF<sub>2</sub>H can be promoted. Depending on applications of the fluorine-containing copolymer, however, it should be paid regard to that contamination by the alkali metal salt must be avoided.

[0069] The fluorine-containing copolymer obtained by the polymerization may be subjected to a fluorination treatment, or may not be subjected to a fluorination treatment. From the viewpoint of avoiding temporal and economic burdens, it is preferable that the fluorine-containing copolymer is not subjected to a fluorination treatment. The fluorination treatment can be carried out by bringing the fluorine-containing copolymer subjected to no fluorination treatment into contact with a fluorine-containing compound. The fluorination treatment can convert thermally unstable functional groups of the fluorine-containing copolymer, such as the carbonyl group-containing terminal groups and —CH<sub>2</sub>OH, and thermally relatively stable functional groups thereof, such as —CF<sub>2</sub>H, to thermally very stable —CF<sub>3</sub>. Resultantly, the total number of the carbonyl group-containing terminal groups and —CH<sub>2</sub>OH of the fluorine-containing copolymer can easily be regulated in the above-mentioned range.

[0070] The fluorine-containing compound is not limited, but includes fluorine radical sources generating fluorine radicals under the fluorination treatment condition. The fluorine radical sources include F<sub>2</sub> gas, CoF<sub>3</sub>, AgF<sub>2</sub>, UF<sub>6</sub>, OF<sub>2</sub>, N<sub>2</sub>F<sub>2</sub>, CF<sub>3</sub>OF, halogen fluorides (for example, IF<sub>5</sub> and ClF<sub>3</sub>).

[0071] The fluorine radical source such as F<sub>2</sub> gas may be, for example, one having a concentration of 100%, but from the viewpoint of safety, the fluorine radical source is preferably mixed with an inert gas and diluted therewith to 5 to 50% by mass, and then used; and it is more preferably to be diluted to 15 to 30% by mass. The inert gas includes nitrogen gas, helium gas and argon gas, but from the viewpoint of the economic efficiency, nitrogen gas is preferred.

[0072] The condition of the fluorination treatment is not limited, and the fluorine-containing copolymer in a melted state may be brought into contact with the fluorine-containing compound, but the fluorination treatment can be carried out usually at a temperature of not higher than the melting point of the fluorine-containing copolymer, preferably at 20 to 220° C. and more preferably at 100 to 200° C. The fluorination treatment is carried out usually for 1 to 30 hours and preferably 5 to 25 hours. The fluorination treatment is preferred which brings the fluorine-containing copolymer having been subjected to no fluorination treatment into contact with fluorine gas (F<sub>2</sub> gas).

[0073] A composition may be obtained by mixing the fluorine-containing copolymer of the present disclosure and as required, other components. The other components

include fillers, plasticizers, processing aids, mold release agents, pigments, flame retarders, lubricants, light stabilizers, weathering stabilizers, electrically conductive agents, antistatic agents, ultraviolet absorbers, antioxidants, foaming agents, perfumes, oils, softening agents and dehydrofluorination agents.

[0074] Examples of the fillers include silica, kaolin, clay, organo clay, talc, mica, alumina, calcium carbonate, calcium terephthalate, titanium oxide, calcium phosphate, calcium fluoride, lithium fluoride, crosslinked polystyrene, potassium titanate, carbon, boron nitride, carbon nanotube and glass fiber. The electrically conductive agents include carbon black. The plasticizers include dioctyl phthalate and pentaerythritol. The processing aids include carnauba wax, sulfone compounds, low molecular weight polyethylene and fluorine-based auxiliary agents. The dehydrofluorination agents include organic oniums and amidines.

[0075] Then, the other components may be other polymers other than the above-mentioned fluorine-containing copolymer. The other polymers include fluororesins other than the above fluorine-containing copolymer, fluoroelastomers and non-fluorinated polymers.

[0076] A method of producing the above composition includes a method in which the fluorine-containing copolymer and other components are dry mixed, and a method in which the fluorine-containing copolymer and other components are previously mixed by a mixer, and then, melt kneaded by a kneader, a melt extruder or the like.

[0077] The fluorine-containing copolymer of the present disclosure or the above-mentioned composition can be used as a processing aid, a forming material or the like, but it is suitable to use that as a forming material. Then, aqueous dispersions, solutions and suspensions of the fluorine-containing copolymer of the present disclosure, and the copolymer/solvent-based materials can also be utilized; and these can be used for application of coating materials, encapsulation, impregnation, and casting of films. However, since the fluorine-containing copolymer of the present disclosure has the above-mentioned properties, it is preferable to use the copolymer as the forming material.

[0078] The fluorine-containing copolymer of the present disclosure or the above composition can be used also as a powder coating material. The powder coating material can be produced, for example, by producing a copolymer, thereafter compressing the obtained copolymer into a sheet shape by a roll, and crushing by a crusher and classifying. The average particle size of the powder coating material (copolymer particles) is preferably 10 to 1,000 μm. The powder coating material is usually applied by being applied on a coating target and thereafter being heated and baked to form a film. The coating film thus applied can be used for various applications as corrosion-resistant linings and the like.

[0079] Formed articles may be obtained by forming the fluorine-containing copolymer of the present disclosure or the above composition.

[0080] A method of forming the fluorine-containing copolymer or the composition is not limited, and includes injection molding, extrusion forming, compression molding, blow molding, transfer molding, rotomolding and rotolining molding. As the forming method, among these, preferable are extrusion forming, compression molding, injection molding and transfer molding; from the viewpoint of being able to produce forming articles in a high productivity, more preferable are extrusion forming and transfer molding, and

still more preferable is extrusion forming. That is, it is preferable that formed articles are extrusion formed articles, compression molded articles, injection molded articles or transfer molded articles; and from the viewpoint of being able to produce molded articles in a high productivity, being extrusion formed articles or transfer molded articles is more preferable, and being extrusion formed articles is still more preferable. Beautiful formed articles or molded articles can be obtained by forming or molding the fluorine-containing copolymer by an extrusion forming method or a transfer molding method.

**[0081]** Formed articles containing the fluorine-containing copolymer of the present disclosure may be, for example, nuts, bolts, joints, films, bottles, gaskets, electric wire coatings, tubes, hoses, pipes, valves, sheets, seals, packings, tanks, rollers, containers, cocks, connectors, filter housings, filter cages, flowmeters, pumps, wafer carriers, and wafer boxes.

**[0082]** The fluorine-containing copolymer of the present disclosure, the above composition and the above formed articles can be used, for example, in the following applications.

**[0083]** Food packaging films, and members for liquid transfer for food production apparatuses, such as lining materials of fluid transfer lines, packings, sealing materials and sheets, used in food production processes;

**[0084]** chemical stoppers and packaging films for chemicals, and members for chemical solution transfer, such as lining materials of liquid transfer lines, packings, sealing materials and sheets, used in chemical production processes; inner surface lining materials of chemical solution tanks and piping of chemical plants and semiconductor factories; members for fuel transfer, such as O (square) rings, tubes, packings, valve stem materials, hoses and sealing materials, used in fuel systems and peripheral equipment of automobiles, and such as hoses and sealing materials, used in ATs of automobiles;

**[0085]** members used in engines and peripheral equipment of automobiles, such as flange gaskets of carburetors, shaft seals, valve stem seals, sealing materials and hoses, and other vehicular members such as brake hoses, hoses for air conditioners, hoses for radiators, and electric wire coating materials;

**[0086]** members for chemical transfer for semiconductor production apparatuses, such as O (square) rings, tubes, packings, valve stem materials, hoses, sealing materials, rolls, gaskets, diaphragms and joints;

**[0087]** members for coating and inks, such as coating rolls, hoses and tubes, for coating facilities, and containers for inks; members for food and beverage transfer, such as tubes, hoses, belts, packings and joints for food and beverage, food packaging materials, and members for glass cooking appliances; members for waste liquid transport, such as tubes and hoses for waste transport;

**[0088]** members for high-temperature liquid transport, such as tubes and hoses for high-temperature liquid transport;

**[0089]** members for steam piping, such as tubes and hoses for steam piping;

**[0090]** corrosionproof tapes for piping, such as tapes wound on piping of decks and the like of ships;

**[0091]** various coating materials, such as electric wire coating materials, optical fiber coating materials, and transparent front side coating materials installed on the

light incident side and back side lining materials of photoelectromotive elements of solar cells;

**[0092]** diaphragms and sliding members such as various types of packings of diaphragm pumps;

**[0093]** films for agriculture, and weathering covers for various kinds of roof materials, sidewalls and the like;

**[0094]** interior materials used in the building field, and coating materials for glasses such as non-flammable fireproof safety glasses; and lining materials for laminate steel sheets used in the household electric field.

**[0095]** The fuel transfer members used in fuel systems of automobiles further include fuel hoses, filler hoses and evaporator hoses. The above fuel transfer members can also be used as fuel transfer members for gasoline additive-containing fuels, resultant to sour gasoline, resultant to alcohols, and resultant to methyl tertiary butyl ether and amines and the like.

**[0096]** The above chemical stoppers and packaging films for chemicals have excellent chemical resistance to acids and the like. The above chemical solution transfer members also include corrosionproof tapes wound on chemical plant pipes.

**[0097]** The above formed articles also include vehicular radiator tanks, chemical solution tanks, bellows, spacers, rollers and gasoline tanks, waste solution transport containers, high-temperature liquid transport containers and fishery and fish farming tanks.

**[0098]** The above formed articles further include members used for vehicular bumpers, door trims and instrument panels, food processing apparatuses, cooking devices, water- and oil-repellent glasses, illumination-related apparatuses, display boards and housings of OA devices, electrically illuminated billboards, displays, liquid crystal displays, cell phones, printed circuit boards, electric and electronic components, sundry goods, dust bins, bathtubs, unit baths, ventilating fans, illumination frames and the like.

**[0099]** Since formed articles containing the fluorine-containing copolymer of the present disclosure are excellent in the 45° C. abrasion resistance, the 82.5° C. high-temperature rigidity, the 115° C. tensile creep resistance, the 50° C. creep resistance, the durability to repeated loads, the chemical solution low permeation, and the ductility to tensile force applied at 100° C., the formed articles can suitably be utilized for tubes, pipes, sheets or the like.

**[0100]** The formed articles containing the fluorine-containing copolymer of the present disclosure can suitably be utilized as members to be compressed such as gaskets and packings. The members to be compressed of the present disclosure may be gaskets or packings.

**[0101]** The size and shape of the members to be compressed of the present disclosure may suitably be set according to applications, and are not limited. The shape of the members to be compressed of the present disclosure may be, for example, annular. The members to be compressed of the present disclosure may also have, in plan view, a circular shape, an elliptic shape, a corner-rounded square or the like, and may be a shape having a throughhole in the central portion thereof.

**[0102]** It is preferable that the members to be compressed of the present disclosure are used as members constituting non-aqueous electrolyte batteries. The members to be compressed of the present disclosure are especially suitable as members to be used in the state of contacting with non-aqueous electrolytes in non-aqueous electrolyte batteries. That is, the members to be compressed of the present disclosure may also be ones having a liquid-contact surface with a non-aqueous electrolyte in the non-aqueous electrolyte batteries.



[0103] The non-aqueous electrolyte batteries are not limited as long as being batteries having a non-aqueous electrolyte, and examples thereof include lithium ion secondary batteries and lithium ion capacitors. Members constituting the non-aqueous electrolyte batteries include sealing members and insulating members.

[0104] For the non-aqueous electrolyte, one or two or more of well-known solvents can be used such as propylene carbonate, ethylene carbonate, butylene carbonate,  $\gamma$ -butyrolactone, 1,2-dimethoxyethane, 1,2-diethoxyethane, dimethyl carbonate, diethyl carbonate and ethyl methyl carbonate. The non-aqueous electrolyte batteries may further have an electrolyte. The electrolyte is not limited, but may be  $\text{LiClO}_4$ ,  $\text{LiAsF}_6$ ,  $\text{LiPF}_6$ ,  $\text{LiBF}_4$ ,  $\text{LiCl}$ ,  $\text{LiBr}$ ,  $\text{CH}_3\text{SO}_3\text{Li}$ ,  $\text{CF}_3\text{SO}_3\text{Li}$ , cesium carbonate and the like.

[0105] The members to be compressed of the present disclosure can suitably be utilized, for example, as sealing members such as sealing gaskets and sealing packings, and insulating members such as insulating gaskets and insulating packings.

[0106] The sealing members are members to be used for preventing leakage of a liquid or a gas, or penetration of a liquid or a gas from the outside. The insulating members are members to be used for insulating electricity. The members to be compressed of the present disclosure may also be members to be used for the purpose of both of sealing and insulation.

[0107] The members to be compressed of the present disclosure can suitably be used as sealing members for non-aqueous electrolyte solution batteries and insulating members for non-aqueous electrolyte solution batteries. Further, the members to be compressed of the present disclosure, due to containing the above fluorine-containing copolymer, have the excellent insulating property. Therefore, in the case of using the members to be compressed of the present disclosure as insulating members, the member firmly adheres to two or more electrically conductive members and prevents short circuit over a long term.

[0108] The fluorine-containing copolymer of the present disclosure can suitably be utilized as a material for forming electric wire coatings. Coated electric wires having a coating layer containing the fluorine-containing copolymer of the present disclosure, since exhibiting almost no fluctuation in the outer diameter, are excellent in electric properties.

[0109] The coated electric wire has a core wire, and the coating layer installed on the periphery of the core wire and containing the fluorine-containing copolymer of the present disclosure. For example, an extrusion formed article made by melt extruding the fluorine-containing copolymer in the present disclosure on a core wire can be made into the coating layer. The coated electric wires are suitable for LAN cables (Ethernet Cables), high-frequency transmission cables, flat cables and heat-resistant cables and the like, and particularly, for transmission cables such as LAN cables (Ethernet Cables) and high-frequency transmission cables.

[0110] As a material for the core wire, for example, a metal conductor material such as copper or aluminum can be used. The core wire is preferably one having a diameter of 0.02 to 3 mm. The diameter of the core wire is more preferably 0.04 mm or larger, still more preferably 0.05 mm or larger and especially preferably 0.1 mm or larger. The diameter of the core wire is more preferable 2 mm or smaller.

[0111] With regard to specific examples of the core wire, there may be used, for example, AWG (American Wire Gauge)-46 (solid copper wire of 40  $\mu\text{m}$  in diameter), AWG-26 (solid copper wire of 404  $\mu\text{m}$  in diameter), AWG-24

(solid copper wire of 510  $\mu\text{m}$  in diameter), and AWG-22 (solid copper wire of 635  $\mu\text{m}$  in diameter).

[0112] The coating layer is preferably one having a thickness of 0.1 to 3.0 mm. It is also preferable that the thickness of the coating layer is 2.0 mm or smaller.

[0113] The high-frequency transmission cables include coaxial cables. The coaxial cables generally have a structure configured by laminating an inner conductor, an insulating coating layer, an outer conductor layer and a protective coating layer in order from the core part to the peripheral part. A formed article containing the fluorine-containing copolymer of the present disclosure can suitably be utilized as the insulating coating layer containing the fluorine-containing copolymer. The thickness of each layer in the above structure is not limited, but is usually: the diameter of the inner conductor is approximately 0.1 to 3 mm; the thickness of the insulating coating layer is approximately 0.3 to 3 mm; the thickness of the outer conductor layer is approximately 0.5 to 10 mm; and the thickness of the protective coating layer is approximately 0.5 to 2 mm.

[0114] Alternatively, the coating layer may be one containing cells, and is preferably one in which cells are homogeneously distributed.

[0115] The average cell size of the cells is not limited, but is, for example, preferably 60  $\mu\text{m}$  or smaller, more preferably 45  $\mu\text{m}$  or smaller, still more preferably 35  $\mu\text{m}$  or smaller, further still more preferably 30  $\mu\text{m}$  or smaller, especially preferable 25  $\mu\text{m}$  or smaller and further especially preferably 23  $\mu\text{m}$  or smaller. Then, the average cell size is preferably 0.1  $\mu\text{m}$  or larger and more preferably 1  $\mu\text{m}$  or larger. The average cell size can be determined by taking an electron microscopic image of an electric wire cross section, calculating the diameter of each cell and averaging the diameters.

[0116] The foaming ratio of the coating layer may be 20% or higher, and is more preferably 30% or higher, still more preferably 33% or higher and further still more preferably 35% or higher. The upper limit is not limited, but is, for example, 80%. The upper limit of the foaming ratio may be 60%. The foaming ratio is a value determined as ((the specific gravity of an electric wire coating material—the specific gravity of the coating layer)/(the specific gravity of the electric wire coating material) $\times 100$ ). The foaming ratio can suitably be regulated according to applications, for example, by regulation of the amount of a gas, described later, to be injected in an extruder, or by selection of the kind of a gas dissolving.

[0117] Alternatively, the coated electric wire may have another layer between the core wire and the coating layer, and may further have another layer (outer layer) on the periphery of the coating layer. In the case where the coating layer contains cells, the electric wire of the present disclosure may be of a two-layer structure (skin-foam) in which a non-foaming layer is inserted between the core wire and the coating layer, a two-layer structure (foam-skin) in which a non-foaming layer is coated as the outer layer, or a three-layer structure (skin-foam-skin) in which a non-foaming layer is coated as the outer layer of the skin-foam structure. The non-foaming layer is not limited, and may be a resin layer composed of a resin, such as a TFE/HFP-based copolymer, a TFE/PAVE copolymer, a TFE/ethylene-based copolymer, a vinylidene fluoride-based polymer, a polyolefin resin such as polyethylene [PE], or polyvinyl chloride [PVC].

[0118] The coated electric wire can be produced, for example, by using an extruder, heating the fluorine-contain-

ing copolymer, extruding the fluorine-containing copolymer in a melt state on the core wire to thereby form the coating layer.

[0119] In formation of a coating layer, by heating the fluorine-containing copolymer and introducing a gas in the fluorine-containing copolymer in a melt state, the coating layer containing cells can be formed. As the gas, there can be used, for example, a gas such as chlorodifluoromethane, nitrogen or carbon dioxide, or a mixture thereof. The gas may be introduced as a pressurized gas in the heated fluorine-containing copolymer, or may be generated by mingling a chemical foaming agent in the fluorine-containing copolymer. The gas dissolves in the fluorine-containing copolymer in a melt state.

[0120] Then, the fluorine-containing copolymer of the present disclosure can suitably be utilized as a material for products for high-frequency signal transmission.

[0121] The products for high-frequency signal transmission are not limited as long as being products to be used for transmission of high-frequency signals, and include (1) formed boards such as insulating boards for high-frequency circuits, insulating materials for connection parts and printed circuit boards, (2) formed articles such as bases of high-frequency vacuum tubes and antenna covers, and (3) coated electric wires such as coaxial cables and LAN cables. The products for high-frequency signal transmission can suitably be used in devices utilizing microwaves, particularly microwaves of 3 to 30 GHz, in satellite communication devices, cell phone base stations, and the like.

[0122] In the products for high-frequency signal transmission, the fluorine-containing copolymer of the present disclosure can suitably be used as insulators in that the dielectric loss tangent is low.

[0123] As the (1) formed boards, printed wiring boards are preferable in that the good electric property is provided. The printed wiring boards are not limited, but examples thereof include printed wiring boards of electronic circuits for cell phones, various computers, communication devices and the like. As the (2) formed articles, antenna covers are preferable in that the dielectric loss is low.

[0124] The fluorine-containing copolymer of the present disclosure can suitably be utilized for films.

[0125] The films of the present disclosure are useful as release films. The release films can be produced by forming the fluorine-containing copolymer of the present disclosure by melt extrusion, calendering, press molding, casting or the like. From the viewpoint that uniform thin films can be obtained, the release films can be produced by melt extrusion.

[0126] The films of the present disclosure can be applied to roll surfaces used in OA devices. Then, the fluorine-containing copolymer of the present disclosure is formed into needed shapes by extrusion forming, compression molding, press molding or the like to be formed into sheet-shapes, filmy shapes or tubular shapes, and can be used as surface materials for OA device rolls, OA device belts or the like. Thin-wall tubes and films can be produced particularly by a melt extrusion forming method.

[0127] The fluorine-containing copolymer of the present disclosure can suitably be utilized also for tubes, bottles and the like.

[0128] So far, embodiments have been described, but it is to be understood that various changes and modifications of patterns and details may be made without departing from the subject matter and the scope of the claims.

[0129] <1> According to a first aspect of the present disclosure, there is provided a fluorine-containing

copolymer, comprising: tetrafluoroethylene unit; hexafluoropropylene unit; and fluoro(alkyl vinyl ether) unit,

[0130] wherein the copolymer has: a content of hexafluoropropylene unit of 9.7 to 10.3% by mass with respect to the whole of the monomer units;

[0131] a content of fluoro(alkyl vinyl ether) unit of 0 to 0.3% by mass with respect to the whole of the monomer units;

[0132] a melt flow rate at 372° C. of 0.80 to 1.30 g/10 min; and

[0133] a total number of a carbonyl group-containing terminal group,  $-\text{CF}=\text{CF}_2$  and  $-\text{CH}_2\text{OH}$  per  $10^6$  main-chain carbon atoms of 50 or less.

[0134] <2> According to a second aspect of the present disclosure, there is provided the fluorine-containing copolymer according to the first aspect, wherein the copolymer has a content of hexafluoropropylene unit of 9.8 to 10.2% by mass with respect to the whole of the monomer units.

[0135] <3> According to a third aspect of the present disclosure, there is provided the fluorine-containing copolymer according to the first or second aspect, wherein the copolymer has a content of fluoro(alkyl vinyl ether) unit of 0 to 0.1% by mass with respect to the whole of the monomer units.

[0136] <4> According to a fourth aspect of the present disclosure, there is provided the fluorine-containing copolymer according to any one of the first to third aspects, wherein the copolymer has a melt flow rate at 372° C. of 0.90 to 1.28 g/10 min.

[0137] <5> According to a fifth aspect of the present disclosure, there is provided the fluorine-containing copolymer according to any one of the first to fourth aspects, wherein the copolymer has the number of  $-\text{CF}_2\text{H}$  per  $10^6$  main-chain carbon atoms of 50 or more.

[0138] <6> According to a sixth aspect of the present disclosure, there is provided the fluorine-containing copolymer according to any one of the first to fifth aspects, wherein the fluoro(alkyl vinyl ether) unit is perfluoro(propyl vinyl ether) unit.

[0139] <7> According to a seventh aspect of the present disclosure, there is provided an extrusion formed article comprising the fluorine-containing copolymer according to any one of the first to sixth aspects.

[0140] <8> According to an eighth aspect of the present disclosure, there is provided a transfer molded article comprising the fluorine-containing copolymer according to any one of the first to sixth aspects.

[0141] <9> According to a ninth aspect of the present disclosure, there is provided a powder coating material comprising the fluorine-containing copolymer according to any one of the first to sixth aspects.

[0142] <10> According to a tenth aspect of the present disclosure, there is provided a coated electric wire comprising a coating layer comprising the fluorine-containing copolymer according to any one of the first to sixth aspects.

[0143] <11> According to an eleventh aspect of the present disclosure, there is provided a formed article comprising the fluorine-containing copolymer according to any one of the first to sixth aspects, wherein the formed article is a tube, a pipe or a sheet.

## EXAMPLES

[0144] Then, the embodiments of the present disclosure will be described by way of Examples, but the present disclosure is not any more limited only to these Examples.

[0145] Each numerical value in Examples was measured by the following methods.

(Content of Monomer Units)

[0146] The content of each monomer unit of the fluorine-containing copolymer was measured by an NMR analyzer (for example, manufactured by Bruker BioSpin GmbH, AVANCE 300, high-temperature probe), or an infrared absorption spectrometer (manufactured by PerkinElmer, Inc., Spectrum One)

(Melt Flow Rate (MFR))

[0147] The MFR of the fluorine-containing copolymer was determined by using a Melt Indexer G-01 (manufac-

N of the functional group per  $1 \times 10^6$  carbon atoms in the sample was calculated according to the following formula (A).

$$N = I \times K / t \quad (A)$$

[0150] I: absorbance

[0151] K: correction factor

[0152] t: thickness of film (mm)

[0153] Regarding the functional groups in Examples, for reference, the absorption frequency, the molar absorption coefficient and the correction factor are shown in Table 2. The molar absorption coefficients are those determined from FT-IR measurement data of low molecular model compounds.

[Table 2]

TABLE 2

| Functional Group                       | Absorption Frequency (cm <sup>-1</sup> ) | Molar Extinction Coefficient (l/cm/mol) | Correction Factor | Model Compound                                    |
|----------------------------------------|------------------------------------------|-----------------------------------------|-------------------|---------------------------------------------------|
| —COF                                   | 1883                                     | 600                                     | 388               | C <sub>7</sub> F <sub>15</sub> COF                |
| —COOH free                             | 1815                                     | 530                                     | 439               | H(CF <sub>2</sub> ) <sub>6</sub> COOH             |
| —COOH bonded                           | 1779                                     | 530                                     | 439               | H(CF <sub>2</sub> ) <sub>6</sub> COOH             |
| —COOCH <sub>3</sub>                    | 1795                                     | 680                                     | 342               | C <sub>7</sub> F <sub>15</sub> COOCH <sub>3</sub> |
| —CONH <sub>2</sub>                     | 3436                                     | 506                                     | 460               | C <sub>7</sub> H <sub>15</sub> CONH <sub>2</sub>  |
| —CH <sub>2</sub> OH <sub>2</sub> , —OH | 3648                                     | 104                                     | 2236              | C <sub>7</sub> H <sub>15</sub> CH <sub>2</sub> OH |
| —CF=CF <sub>2</sub>                    | 1795                                     | 635                                     | 366               | CF <sub>2</sub> =CF <sub>2</sub>                  |

tured by Toyo Seiki Seisaku-sho, Ltd.), and making the polymer to flow out from a die of 2 mm in inner diameter and 8 mm in length at 372° C. under a load of 5 kg and measuring the mass (g/10 min) of the polymer flowing out per 10 min, according to ASTM D1238.

(The Number of —CF<sub>2</sub>H)

[0148] The number of —CF<sub>2</sub>H groups of the fluorine-containing copolymer was determined from a peak integrated value of the —CF<sub>2</sub>H group acquired in a <sup>19</sup>F-NMR measurement using a nuclear magnetic resonance spectrometer AVANCE-300 (manufactured by Bruker BioSpin GmbH) and set at a measurement temperature of (the melting point of the polymer+20° C.)

(The Numbers of —COOH, —COOCH<sub>3</sub>, —CH<sub>2</sub>OH, —COF, —CF=CF<sub>2</sub> and —CONH<sub>2</sub>)

[0149] A dried powder or pellets obtained in each of Examples and Comparative Examples were molded by cold press to prepare a film of 0.25 to 0.3 mm in thickness. The film was 40 times scanned by a Fourier transform infrared spectrometer [FT-IR (Spectrum One, manufactured by PerkinElmer, Inc.)] and analyzed to obtain an infrared absorption spectrum. The obtained infrared absorption spectrum was compared with an infrared absorption spectrum of an already known film to determine the kinds of terminal groups. Further, from an absorption peak of a specific functional group emerging in a difference spectrum between the obtained infrared absorption spectrum and the infrared absorption spectrum of the already known film, the number

(The Number of —OC(=O)O—R (Carbonate Groups))

[0154] Analysis of the number of —OC(=O)O—R (carbonate groups) was carried out by a method described in International Publication No. WO2019/220850. The number of —OC(=O)O—R (carbonate groups) was calculated as in the calculation method of the number of functional groups, N, except for setting the absorption frequency at 1,817 cm<sup>-1</sup>, the molar extinction coefficient at 170 (l/cm/mol) and the correction factor at 1,426.

(Melting Point)

[0155] The fluorine-containing copolymer was heated, as a first temperature raising step at a temperature-increasing rate of 10° C./min from 200° C. to 350° C., then cooled at a cooling rate of 10° C./min from 350° C. to 200° C., and then again heated, as second temperature raising step, at a temperature-increasing rate of 10° C./min from 200° C. to 350° C. by using a differential scanning calorimeter (trade name: X-DSC7000, manufactured by Hitachi High-Tech Science Corp.); and the melting point of the fluorine-containing copolymer was determined from a melting curve peak observed in the second temperature raising step.

## Comparative Example 1

[0156] 40.25 kg of deionized water and 0.079 kg of methanol were fed in a 174 L-volume autoclave with a stirrer, and the autoclave inside was sufficiently vacuumized and replaced with nitrogen. Thereafter, the autoclave inside was vacuum deaerated, and in the autoclave put in a vacuum

state, 40.25 kg of HFP was fed; and the autoclave was heated to 25.5° C. Then, TFE was fed until the internal pressure of the autoclave became 0.832 MPa; and then, 1.25 kg of an 8-mass % di( $\omega$ -hydroperfluorohexanoyl) peroxide solution (hereinafter, abbreviated to DHP) was fed in the autoclave to initiate polymerization. The internal pressure of the autoclave at the initiation of the polymerization was set at 0.893 MPa, and by continuously adding TFE, the set pressure was made to be held. After 1.5 hours from the polymerization initiation, 0.079 kg of methanol was additionally fed. After 2 hours and 4 hours from the polymerization initiation, 1.25 kg of DHP was additionally fed, and the internal pressure was lowered by 0.002 MPa, respectively; after 6 hours therefrom, 0.96 kg thereof was fed and the internal pressure was lowered by 0.002 MPa. Hereafter, 0.25 kg of DHP was fed at every 2 hours until the reaction finished, and at every time, the internal pressure was lowered by 0.002 MPa.

[0157] Then, at each time point when the amount of TFE additionally fed reached 6.0 kg and 18.1 kg, 0.079 kg of methanol was additionally fed in the autoclave. Then, when the amount of TFE additionally fed reached 40.25 kg, the polymerization was made to finish. After the finish of the polymerization, unreacted TFE and HFP were discharged to thereby obtain a wet powder. Then, the wet powder was washed with pure water, and thereafter dried at 150° C. for 10 hours to thereby obtain 44.9 kg of a dry powder.

[0158] The obtained powder was melt extruded at 370° C. by a screw extruder (trade name: PCM46, manufactured by Ikegai Corp.) to thereby obtain pellets of a copolymer. By using the obtained pellets, the above physical properties were measured by the methods described above. The results are shown in Table 3.

#### Comparative Example 2

[0159] 40.25 kg of deionized water and 0.047 kg of methanol were fed in a 174 L-volume autoclave with a stirrer, and the autoclave inside was sufficiently vacuumized and replaced with nitrogen. Thereafter, the autoclave inside was vacuum deaerated, and in the autoclave put in a vacuum state, 40.25 kg of HFP was fed; and the autoclave was heated to 30.0° C. Then, TFE was fed until the internal pressure of the autoclave became 0.914 MPa; and then, 0.63 kg of an 8-mass % di( $\omega$ -hydroperfluorohexanoyl) peroxide solution (hereinafter, abbreviated to DHP) was fed in the autoclave to initiate polymerization. The internal pressure of the autoclave at the initiation of the polymerization was set at 0.914 MPa, and by continuously adding TFE, the set pressure was made to be held. After 1.5 hours from the polymerization initiation, 0.047 kg of methanol was additionally fed. After 2 hours and after 4 hours from the polymerization initiation, 0.63 kg of DHP was additionally fed, and the internal pressure was lowered by 0.001 MPa, respectively; after 6 hours therefrom, 0.48 kg thereof was fed and the internal pressure was lowered by 0.001 MPa. Hereafter, 0.13 kg of DHP was fed at every 2 hours until the reaction finished, and at the every time, the internal pressure was lowered by 0.001 MPa.

[0160] Then, at each time point when the amount of TFE additionally fed reached 6.0 kg and 18.1 kg, 0.047 kg of methanol was additionally fed in the autoclave. Then, when the amount of TFE additionally fed reached 40.25 kg, the polymerization was made to finish. After the finish of the polymerization, unreacted TFE and HFP were discharged to thereby obtain a wet powder. Then, the wet powder was washed with pure water, and thereafter dried at 150° C. for 10 hours to thereby obtain 45.5 kg of a dry powder.

[0161] The obtained powder was melt extruded at 370° C. by a screw extruder (trade name: PCM46, manufactured by Ikegai Corp.) to thereby obtain pellets of a copolymer. By using the obtained pellets, the above physical properties were measured by the methods described above. The results are shown in Table 3.

#### Comparative Example 3

[0162] Copolymer pellets were obtained as in Comparative Example 2, except for changing the amount of methanol fed before the polymerization initiation to 0.115 kg, changing the each amount of methanol dividedly additionally fed after the polymerization initiation to 0.115 kg, and changing the each set pressure in the autoclave inside before and after the polymerization initiation to 0.928 MPa. By using the obtained pellets, the above physical properties were measured by the methods described above. The results are shown in Table 3.

#### Comparative Example 4

[0163] Copolymer pellets were obtained as in Comparative Example 2, except for changing the amount of methanol fed before the polymerization initiation to 0.004 kg, changing the each amount of methanol dividedly additionally fed after the polymerization initiation to 0.004 kg, and changing the each set pressure in the autoclave inside before and after the polymerization initiation to 0.926 MPa. By using the obtained pellets, the above physical properties were measured by the methods described above. The results are shown in Table 3.

#### Comparative Example 5

[0164] 40.25 kg of deionized water and 0.042 kg of methanol were fed in a 174 L-volume autoclave with a stirrer, and the autoclave inside was sufficiently vacuumized and replaced with nitrogen. Thereafter, the autoclave inside was vacuum deaerated, and in the autoclave put in a vacuum state, 40.25 kg of HFP and 0.32 kg of PPVE were fed; and the autoclave was heated to 30.0° C. Then, TFE was fed until the internal pressure of the autoclave became 0.928 MPa; and then, 0.63 kg of an 8-mass % di( $\omega$ -hydroperfluorohexanoyl) peroxide solution (hereinafter, abbreviated to DHP) was fed in the autoclave to initiate polymerization. The internal pressure of the autoclave at the initiation of the polymerization was set at 0.928 MPa, and by continuously adding TFE, the set pressure was made to be held. After 1.5 hours from the polymerization initiation, 0.042 kg of methanol was additionally fed. After 2 hours and 4 hours from the polymerization initiation, 0.63 kg of DHP was additionally fed, and the internal pressure was lowered by 0.001 MPa, respectively; after 6 hours therefrom, 0.48 kg thereof was fed and the internal pressure was lowered by 0.001 MPa. Hereafter, 0.13 kg of DHP was fed at every 2 hours until the reaction finished, and at the every time, the internal pressure was lowered by 0.001 MPa.

[0165] Then, at each time point when the amount of TFE continuously additionally fed reached 8.1 kg, 16.2 kg and 24.3 kg, 0.09 kg of PPVE was additionally fed. Then, at each time point when the amount of TFE additionally fed reached 6.0 kg and 18.1 kg, 0.042 kg of methanol was additionally fed in the autoclave. Then, when the amount of TFE additionally fed reached 40.25 kg, the polymerization was made to finish. After the finish of the polymerization,

unreacted TFE and HFP were discharged to thereby obtain a wet powder. Then, the wet powder was washed with pure water, and thereafter dried at 150° C. for 10 hours to thereby obtain 45.6 kg of a dry powder.

[0166] The obtained powder was melt extruded at 370° C. by a screw extruder (trade name: PCM46, manufactured by Ikegai Corp.) to thereby obtain pellets of a copolymer. By using the obtained pellets, the above physical properties were measured by the methods described above. The results are shown in Table 3.

#### Comparative Example 6

[0167] 1.65 kg of deionized water was fed in a 6 L-volume autoclave with a stirrer, and the autoclave inside was sufficiently vacuumized and replaced with nitrogen. Thereafter, the autoclave inside was vacuum deaerated, and in the autoclave put in a vacuum state, 1.65 kg of HFP was fed; and the autoclave was heated to 40.0° C. Then, a mixed gas of TFE and HFP in a 91:9 molar ratio was fed until the internal pressure of the autoclave became 1.330 MPa; and then, 4.3 g of a 40-mass % diisopropyl peroxydicarbonate solution was fed in the autoclave to initiate polymerization. The internal pressure of the autoclave at the initiation of the polymerization was set at 1.330 MPa, and by continuously adding the mixed gas of TFE and HFP in a 91:9 molar ratio, the set pressure was made to be held.

[0168] Then, when the amount of the additional gas fed reached 343 g, the polymerization was made to finish. After the finish of the polymerization, unreacted TFE and HFP were discharged to thereby obtain a wet powder. Then, the wet powder was washed with pure water, and thereafter dried at 110° C. for 10 hours and at 140° C. for 5 hours by a hot-air dryer, and then vacuum dried at 140° C. for 24 hours by a vacuum dryer to thereby obtain 280 g of a dry powder.

#### Example 1

[0169] Copolymer pellets were obtained as in Comparative Example 1, except for changing the amount of methanol fed before the polymerization initiation to 0.012 kg, changing the each amount of methanol dividedly additionally fed after the polymerization initiation to 0.012 kg, and changing the each set pressure in the autoclave inside before and after the polymerization initiation to 0.823 MPa. By using the obtained pellets, the above physical properties were measured by the methods described above. The results are shown in Table 3.

#### Example 2

[0170] 40.25 kg of deionized water and 0.026 kg of methanol were fed in a 174 L-volume autoclave with a stirrer, and the autoclave inside was sufficiently vacuumized and replaced with nitrogen. Thereafter, the autoclave inside was vacuum deaerated, and in the autoclave put in a vacuum state, 40.25 kg of HFP and 0.04 kg of PPVE were fed; and the autoclave was heated to 25.5° C. Then, TFE was fed until the internal pressure of the autoclave became 0.823 MPa; and then, 1.25 kg of an 8-mass % di( $\omega$ -hydroperfluorohexanoyl) peroxide solution (hereinafter, abbreviated to DHP) was fed in the autoclave to initiate polymerization. The internal pressure of the autoclave at the initiation of the polymerization was set at 0.823 MPa, and by continuously adding TFE, the set pressure was made to be held. After 1.5 hours from the polymerization initiation, 0.026 kg of methanol was additionally fed. After 2 hours and 4 hours from the polymerization initiation, 1.25 kg of DHP was additionally

fed, and the internal pressure was lowered by 0.002 MPa, respectively; after 6 hours therefrom, 0.96 kg thereof was fed and the internal pressure was lowered by 0.002 MPa. Hereafter, 0.25 kg of DHP was fed at every 2 hours until the reaction finished, and at the every time, the internal pressure was lowered by 0.002 MPa.

[0171] Then, at each time point when the amount of TFE continuously additionally fed reached 8.1 kg, 16.2 kg and 24.3 kg, 0.01 kg of PPVE was additionally fed. Then, at each time point when the amount of TFE additionally fed reached 6.0 kg and 18.1 kg, 0.026 kg of methanol was additionally fed in the autoclave. Then, when the amount of TFE additionally fed reached 40.25 kg, the polymerization was made to finish. After the finish of the polymerization, unreacted TFE and HFP were discharged to thereby obtain a wet powder. Then, the wet powder was washed with pure water, and thereafter dried at 150° C. for 10 hours to thereby obtain 45.1 kg of a dry powder.

[0172] The obtained powder was melt extruded at 370° C. by a screw extruder (trade name: PCM46, manufactured by Ikegai Corp.) to thereby obtain pellets of a copolymer. By using the obtained pellets, the above physical properties were measured by the methods described above. The results are shown in Table 3.

#### Example 3

[0173] Copolymer pellets were obtained as in Comparative Example 2, except for changing the amount of methanol fed before the polymerization initiation to 0.045 kg, changing the each amount of methanol dividedly additionally fed after the polymerization initiation to 0.045 kg, and changing the each set pressure in the autoclave inside before and after the polymerization initiation to 0.928 MPa. By using the obtained pellets, the HFP content was measured by the methods described above. The results are shown in Table 3.

[0174] The obtained pellets were deaerated at 200° C. for 8 hours in an electric furnace, put in a vacuum vibration-type reactor VVD-30 (manufactured by Okawara Mfg. Co. Ltd.), and heated to 200° C. After vacuumizing, F<sub>2</sub> gas diluted to 20% by volume with N<sub>2</sub> gas was introduced to the atmospheric pressure. 0.5 hour after the F<sub>2</sub> gas introduction, vacuumizing was once carried out and F<sub>2</sub> gas was again introduced. Further, 0.5 hour thereafter, vacuumizing was again carried out and F<sub>2</sub> gas was again introduced. Thereafter, while the above operation of the F<sub>2</sub> gas introduction and the vacuumizing was carried out once every 1 hour, the reaction was carried out at a temperature of 200° C. for 8 hours. After the reaction was finished, the reactor interior was replaced sufficiently by N<sub>2</sub> gas to finish the fluorination reaction, whereby pellets were obtained. By using the obtained pellets, the above physical properties were measured by the methods described above. The results are shown in Table 3.

#### Example 4

[0175] Copolymer pellets were obtained as in Comparative Example 2, except for changing the amount of methanol fed before the polymerization initiation to 0.086 kg, changing the each amount of methanol dividedly additionally fed after the polymerization initiation to 0.086 kg, and changing the each set pressure in the autoclave inside before and after the polymerization initiation to 0.923 MPa. By using the obtained pellets, the above physical properties were measured by the methods described above. The results are shown in Table 3.

TABLE 3

|                       | HFP<br>content<br>(% by mass) | PPVE<br>content<br>(% by mass) | MFR<br>(g/10 min) | —CF <sub>2</sub> H<br>(number/C10 <sup>6</sup> ) | —COOH<br>—COF<br>—CH <sub>2</sub> OH<br>(number/C10 <sup>6</sup> ) | —OCOOR<br>(number/C10 <sup>6</sup> ) | Others<br>(number/C10 <sup>6</sup> ) | Melting point<br>(° C.) |
|-----------------------|-------------------------------|--------------------------------|-------------------|--------------------------------------------------|--------------------------------------------------------------------|--------------------------------------|--------------------------------------|-------------------------|
| Comparative Example 1 | 9.4                           | 0                              | 1.24              | 114                                              | 13                                                                 | ND                                   | <6                                   | 270                     |
| Comparative Example 2 | 10.7                          | 0                              | 1.20              | 109                                              | 15                                                                 | ND                                   | <6                                   | 262                     |
| Comparative Example 3 | 10.0                          | 0                              | 1.44              | 132                                              | 14                                                                 | ND                                   | <6                                   | 266                     |
| Comparative Example 4 | 10.1                          | 0                              | 0.60              | 73                                               | 14                                                                 | ND                                   | <6                                   | 264                     |
| Comparative Example 5 | 10.0                          | 0.8                            | 1.15              | 96                                               | 22                                                                 | ND                                   | <6                                   | 260                     |
| Comparative Example 6 | 9.9                           | 0                              | 1.00              | 10                                               | 12                                                                 | 174                                  | <6                                   | 267                     |
| Example 1             | 9.8                           | 0                              | 0.90              | 84                                               | 14                                                                 | ND                                   | <6                                   | 266                     |
| Example 2             | 9.8                           | 0.1                            | 1.06              | 94                                               | 15                                                                 | ND                                   | <6                                   | 266                     |
| Example 3             | 10.0                          | 0                              | 1.10              | <9                                               | <6                                                                 | ND                                   | <6                                   | 266                     |
| Example 4             | 10.2                          | 0                              | 1.28              | 118                                              | 14                                                                 | ND                                   | <6                                   | 265                     |

[0176] The description of “Others (number/C10<sup>6</sup>)” in Table 3 denotes the total number of —COOCH<sub>3</sub>, —CF=CF<sub>2</sub> and —CONH<sub>2</sub>. The description of “<9” in Table 3 means that the number (total number) of —CF<sub>2</sub>H groups was less than 9. The description of “<6” in Table 3 means that the number (total number) of the objective functional groups was less than 6. The description of “ND” in Table 3 means that for the objective functional group, no quantitatively determinable peak could be observed.

[0177] Then, by using the obtained pellets or dry powder, the following properties were evaluated. The results are shown in Table 4.

#### (Abrasion Test)

[0178] By using the pellets or the dry powder and a heat press molding machine, a sheet-shape test piece of approximately 0.2 mm in thickness was prepared and cut out into a test piece of 10 cm×10 cm. The prepared test piece was fixed on a test bench of a Taber abrasion tester (No. 101, Taber type abrasion tester with an option, manufactured by Yasuda Seiki Seisakusho, Ltd.), and the abrasion test was carried out under the conditions of at a test piece surface temperature of 45° C., at a load of 500 g, using an abrasion wheel CS-10 (rotationally polished in 20 rotations with an abrasion paper #240), and at a rotation rate of 60 rpm, using the Taber abrasion tester. The weight of the test piece after 1,000 rotations was measured, and the same test piece was further subjected to the test of 15,300 rotations and thereafter, the weight thereof was measured. The abrasion loss was determined by the following formula.

$$\text{Abrasion loss (mg)} = M1 - M2$$

[0179] M1: the weight of the test piece after the 1,000 rotations (mg)

[0180] M2: the weight of the test piece after the 15,300 rotations (mg)

#### (82.5° C. Load Deflection Rate)

[0181] By using the pellets or the dry powder and a heat press molding machine, a sheet-shape test piece of approximately 2.7 mm in thickness was prepared and cut out into a

test piece of 80×10 mm; and the test piece was heated in an electric furnace at 100° C. for 20 hours. A test was carried out according to a method described in JIS K-K7191-1 except for using the obtained test piece, by a heat distortion tester (manufactured by Yasuda Seiki Seisakusho, Ltd.) under the conditions of at a test temperature of 30 to 150° C., at a temperature-increasing rate of 120° C./h, at a bending stress of 1.8 MPa and the flatwise method. The load deflection rate was determined by the following formula. A sheet low in the load deflection rate at 82.5° C. is excellent in the high-temperature rigidity.

$$\text{Load deflection rate (\%)} = a_2/a_1 \times 100$$

[0182] a<sub>1</sub>: a thickness of the test piece before the test (mm)

[0183] a<sub>2</sub>: an amount deflected at 82.5° C. (mm)

#### (Tensile Creep Test)

[0184] The tensile creep strain was measured by using TMA-7100, manufactured by Hitachi High-Tech Science Corp. By using the pellets or the dry powder and a heat press molding machine, a sheet of approximately 0.1 mm in thickness was prepared, and a sample of 2 mm in width and 22 mm in length was prepared from the sheet. The sample was mounted on measurement jigs with the distance between the jigs of 10 mm. A load was applied on the sample so that the cross-sectional load became 4.30 N/mm<sup>2</sup>, and allowed to stand at 115° C.; and the displacement (mm) of the sample length from the timepoint of 70 min from the test initiation to the timepoint of 825 min from the test initiation was measured, and the proportion (tensile creep strain (%)) of the displacement (mm) of the length to the initial sample length (10 mm) was calculated. A sheet low in the tensile creep strain (%) measured under the condition of at 115° C. for 825 min is hardly elongated even when a tensile load is applied in a very high-temperature environment, being excellent in the high-temperature tensile creep property.

#### (Evaluation of the Creep Resistance)

[0185] The measurement of the creep resistance was carried out according to ASTM D395 or JIS K6262:2013. By

using the pellets or the dry powder and a heat press molding machine, a molded article of 13 mm in outer diameter and 8 mm in height was prepared. The obtained molded article was cut to prepare a test piece of 13 mm in outer diameter and 6 mm in height. The prepared test piece was compressed by using a compression apparatus at ordinary temperature to a compression deformation rate of 25%. The compressed test piece was allowed to stand still in an electric furnace as it was fixed on a compression apparatus, and allowed to stand at 50° C. for 72 hours. The compression apparatus was taken out from the electric furnace, and cooled to room temperature; thereafter, the test piece was dismantled. The recovered test piece was allowed to stand at room temperature for 30 min; thereafter, the height of the recovered test piece was measured, and the extent of recovery was determined by the following formula.

$$\text{Extent of recovery (\%)} = (t_2 - t_1)/t_3 \times 100$$

[0186]  $t_1$ : a height of a spacer (mm)

[0187]  $t_2$ : a height of the test piece dismantled from the compression apparatus (mm)

[0188]  $t_3$ : a compression deformed height (mm)

[0189] In the above test,  $t_1$  was 4.5 mm and  $t_3$  was 1.5 mm. (Tensile Strength after 8,000 Cycles)

[0190] The tensile strength after 8,000 cycles was measured by using a fatigue testing machine MMT-250NV-10, manufactured by Shimadzu Corp. By using the pellets or the dry powder and a heat press molding machine, a sheet of approximately 2.4 mm in thickness was prepared, and a sample in a dumbbell shape (thickness: 2.4 mm, width: 5.0 mm, measuring section length: 22 mm) was prepared by using an ASTM D1708 microdumbbell. The sample was mounted on measuring jigs and the measuring jigs were installed in a state of the sample being mounted in a thermostatic chamber at 110° C. The tensile operation in the uniaxial direction was repeated at a stroke of 0.2 mm and at a frequency of 100 Hz, and there was measured the tensile strength at every tensile operation (tensile strength at the time the stroke was +0.2 mm, unit: N).

(Acetic Acid Permeability)

[0191] By using the pellets or the dry powder and a heat press molding machine, a sheet-shape test piece of approximately 0.2 mm in thickness was prepared. 10 g of acetic acid was put in a test cup (permeation area: 12.56 cm<sup>2</sup>), and the test cup was covered with the sheet-shape test piece; and a PTFE gasket was pinched and fastened to hermetically close the test cup. The sheet-shape test piece was brought into contact with the acetic acid, and held at a temperature of 60° C. for 55 days, and thereafter, the test cup was taken out and allowed to stand at room temperature for 1 hour; thereafter, the amount of the mass lost was measured. The acetic acid permeability (g·cm/m<sup>2</sup>) was determined by the following formula.

Acetic acid permeability (g·cm/m<sup>2</sup>) =

$$\left[ \frac{\text{the amount of the mass lost (g)} \times \text{the thickness of the sheet-shape test piece (cm)}}{\text{the permeation area (m}^2\text{)}} \right]$$

(100° C. Tensile Elongation)

[0192] By using the pellets and a heat press molding machine, a test piece (compression molded) of 2.0 mm in thickness was prepared. A dumbbell shape test piece was punched out from the test piece by using an ASTM V-type dumbbell; and by using the obtained dumbbell shape test piece, the tensile elongation was measured at 100° C. by using an Autograph (AG-I, 300 kN, manufactured by Shimadzu Corp.), according to ASTM D638 under the condition of 50 mm/min.

(Melt Own Deformation Test)

[0193] By using the pellets or the dry powder and a heat press molding machine, a molded article of 13 mm in diameter and approximately 6.5 mm in height was prepared. The obtained molded article was cut into a test piece of 6.3 mm in height. The prepared test piece was put in a SUS-made petri dish, heated at 390° C. for 30 min in an electric furnace, and thereafter water cooled together with the petri dish with the test piece. The diameter of the surface (bottom surface) having contacted with the petri dish of the test piece taken out was measured with slide calipers, and the bottom area increase rate was calculated by the following formula.

$$\text{Bottom area increase rate (\%)} = \left\{ \frac{\text{the test piece bottom area after being heated (mm}^2\text{)} - \text{the test piece bottom area before being heated (mm}^2\text{)}}{\text{the test piece bottom area before being heated (mm}^2\text{)}} \right\} \times 100$$

[0194] A lower bottom area increase rate means that a formed article is more hardly deformed by its own weight in melting. The copolymer giving a formed article low in the bottom area increase rate is excellent in that even in the case where the copolymer is formed into a thick sheet or a large pipe by an extrusion forming method, the formed article in the melt state is hardly deformed and the formed article having a desired shape can be obtained after being cooled and solidified.

(Ammonia Water Permeability)

[0195] By using the pellets or the dry powder and a heat press molding machine, a sheet-shape test piece of approximately 0.1 mm in thickness was prepared. 10 g of a 29% ammonia water was put in a test cup (permeation area: 12.56 cm<sup>2</sup>), and the test cup was covered with the sheet-shape test piece; and a PTFE gasket was pinched and fastened to hermetically close the test cup. The sheet-shape test piece was brought into contact with the ammonia water, and held at a temperature of 37° C. for 28 days, and thereafter, the test cup was taken out and the amount of the mass lost was measured. The ammonia water permeability (mg·cm/m<sup>2</sup>) was determined by the following formula.

Ammonia water permeability (mg·cm/m<sup>2</sup>) =

$$\left[ \frac{\text{the amount of the mass lost (mg)} \times \text{the thickness of the sheet-shape test piece (cm)}}{\text{the permeation area (m}^2\text{)}} \right]$$

(Film Formation)

[0196] By a  $\phi$ 14-mm extruder (manufactured by Imoto Machinery Co. Ltd.), a T die was used to prepare a film. The extrusion conditions were as follows.

[0197] a) Take-up speed: 0.1 m/min

[0198] b) Roll temperature: 120° C.

[0199] c) Film width: 70 mm

[0200] d) Thickness: 1.00 mm

[0201] e) Extrusion condition:

[0202] Cylinder screw diameter=14 mm, a single-screw extruder of L/D=20

Set temperature of the extruder: barrel section C-1 (330° C.), barrel section C-2 (350° C.), barrel section C-3 (360° C.), T die section (365° C.)

[0203] It was visually confirmed that the film was prepared without any problem.

(Tube Formation)

[0204] A tube of 10.0 mm in outer diameter and 1.0 mm in wall thickness was extrusion formed by a 30-mm $\phi$  extruder (manufactured by Tanabe Plastics Machinery Co. Ltd.). The extrusion conditions were as follows.

[0205] a) Die inner diameter: 25 mm

[0206] b) Mandrel outer diameter: 13 mm

[0207] c) Sizing die inner diameter: 10.5 mm

[0208] d) Take-over speed: 0.4 m/min

[0209] e) Outer diameter: 10.0 mm

[0210] f) Wall thickness: 1.0 mm

[0211] g) Extrusion condition:

[0212] Cylinder screw diameter=30 mm, a single-screw extruder of L/D=22

Set temperature of the extruder: barrel section C-1 (350° C.), barrel section C-2 (370° C.), barrel section C-3 (380° C.), head section H-1 (390° C.), die section D-1 (390° C.), die section D-2 (390° C.)

[0213] It was visually confirmed that the tube was prepared without any problem.

(Preparation and Evaluation of the Powder Coating Material)

[0214] The obtained pellets were crushed and then sieved through a No. 18 sieve (manufactured by Tokyo Screen Co. Ltd.) to thereby obtain a copolymer powder. The copolymer powder was applied on a smooth SUS316 substrate (200 mm $\times$ 200 mm $\times$ 5 mm) by an electrostatic coating method, and thereafter vertically suspended in a drying furnace and baked at 350° C. for 30 min to thereby prepare a coating film. It was visually checked that the coating film was made. In Table 4, “good” means that a coating film was made.

TABLE 4

|                       | 45° C.<br>Abrasion<br>loss<br>(mg)      | 82.5° C. Load<br>deflection<br>rate<br>(%) | 115° C.<br>Tensile<br>creep strain<br>(%)                          | Creep<br>resistance<br>evaluation<br>Recovery<br>extent (%) | Tensile<br>strength<br>after 8,000<br>cycles<br>(N) | Acetic acid<br>permeability<br>(g · cm/m <sup>2</sup> ) |
|-----------------------|-----------------------------------------|--------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|---------------------------------------------------------|
| Comparative Example 1 | 31.0                                    | 68%                                        | 1.00                                                               | 21%                                                         | 8.17                                                | 2.6                                                     |
| Comparative Example 2 | 29.0                                    | 80%                                        | 1.54                                                               | 17%                                                         | 6.63                                                | 2.6                                                     |
| Comparative Example 3 | 31.1                                    | 71%                                        | 1.29                                                               | 19%                                                         | 7.37                                                | 2.6                                                     |
| Comparative Example 4 | 26.6                                    | 81%                                        | 0.96                                                               | 19%                                                         | 7.73                                                | 2.7                                                     |
| Comparative Example 5 | 26.2                                    | 87%                                        | 2.33                                                               | 17%                                                         | 7.44                                                | 2.8                                                     |
| Comparative Example 6 | 29.2                                    | 74%                                        | 1.10                                                               | 20%                                                         | 7.68                                                | 3.5                                                     |
| Example 1             | 28.8                                    | 74%                                        | 1.02                                                               | 20%                                                         | 7.88                                                | 2.7                                                     |
| Example 2             | 29.1                                    | 74%                                        | 1.39                                                               | 20%                                                         | 7.77                                                | 2.6                                                     |
| Example 3             | 29.5                                    | 74%                                        | 1.18                                                               | 20%                                                         | 7.51                                                | 2.6                                                     |
| Example 4             | 29.9                                    | 74%                                        | 1.32                                                               | 19%                                                         | 7.20                                                | 2.6                                                     |
|                       | 100° C.<br>Tensile<br>elongation<br>(%) | Own weight<br>deformation<br>test<br>(%)   | NH <sub>3</sub> water<br>permeability<br>(mg · cm/m <sup>2</sup> ) | Film<br>formation                                           | Tube<br>formation                                   | Powder<br>coating<br>material<br>evaluation             |
| Comparative Example 1 | 318                                     | 51                                         | 757                                                                | good                                                        | good                                                | —                                                       |
| Comparative Example 2 | 367                                     | 50                                         | 760                                                                | good                                                        | good                                                | —                                                       |
| Comparative Example 3 | 325                                     | 60                                         | 745                                                                | good                                                        | good                                                | good                                                    |
| Comparative Example 4 | 367                                     | 25                                         | 840                                                                | —                                                           | —                                                   | —                                                       |
| Comparative Example 5 | 397                                     | 48                                         | 794                                                                | good                                                        | good                                                | —                                                       |
| Comparative Example 6 | —                                       | 42                                         | 775                                                                | —                                                           | —                                                   | —                                                       |
| Example 1             | 343                                     | 37                                         | 776                                                                | good                                                        | good                                                | good                                                    |
| Example 2             | 344                                     | 44                                         | 774                                                                | good                                                        | good                                                | good                                                    |
| Example 3             | 344                                     | 46                                         | 700                                                                | good                                                        | good                                                | good                                                    |
| Example 4             | 347                                     | 53                                         | 755                                                                | good                                                        | good                                                | good                                                    |



1. A fluorine-containing copolymer, comprising: tetrafluoroethylene unit; hexafluoropropylene unit; and a fluoro(alkyl vinyl ether) unit,

wherein the copolymer has: a content of hexafluoropropylene unit of 9.7 to 10.3% by mass with respect to the whole of the monomer units;

a content of the fluoro(alkyl vinyl ether) unit of 0 to 0.3% by mass with respect to the whole of the monomer units;

a melt flow rate at 372° C. of 0.80 to 1.30 g/10 min; and

a total number of a carbonyl group-containing terminal group,  $-\text{CF}=\text{CF}_2$  and  $-\text{CH}_2\text{OH}$  per  $10^6$  main-chain carbon atoms of 50 or less.

2. The fluorine-containing copolymer according to claim 1, wherein the copolymer has a content of hexafluoropropylene unit of 9.8 to 10.2% by mass with respect to the whole of the monomer units.

3. The fluorine-containing copolymer according to claim 1, wherein the copolymer has a content of the fluoro(alkyl vinyl ether) unit of 0 to 0.1% by mass with respect to the whole of the monomer units.

4. The fluorine-containing copolymer according to claim 1, wherein the copolymer has a melt flow rate at 372° C. of 0.90 to 1.28 g/10 min.

5. The fluorine-containing copolymer according to claim 1, wherein the copolymer has the number of  $-\text{CF}_2\text{H}$  per  $10^6$  main-chain carbon atoms of 50 or more.

6. The fluorine-containing copolymer according to claim 1, wherein the fluoro(alkyl vinyl ether) unit is perfluoro(propyl vinyl ether) unit.

7. An extrusion formed article comprising the fluorine-containing copolymer according to claim 1.

8. A transfer molded article comprising the fluorine-containing copolymer according to claim 1.

9. A powder coating material comprising the fluorine-containing copolymer according to claim 1.

10. A coated electric wire comprising a coating layer comprising the fluorine-containing copolymer according to claim 1.

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